

[Dashboard](#) / [My courses](#) / [036026 - Kine., Dyn. Control of Robots - Spring 2025](#) / [Exams](#)
/ [Exam A question 1 \(to be done using the Safe Exam Browser\)](#) / [Preview](#)

Started on Thursday, 19 June 2025, 9:28 PM

State Finished

Completed on Thursday, 19 June 2025, 9:30 PM

Time taken 1 min 18 secs

Grade 15.00 out of 15.00 (100%)

Question 1

Correct

Mark 3.00 out of 3.00

For the vector

$$a = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

the matrix $\hat{a}$ is defined as

$$\begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix}.$$

Using Poincot's theorem, the screw coordinates of a wrench $F = (f, \tau) \in \mathbb{R}^6$ are defined as:

1. Pitch:

$$h = \frac{f^T \tau}{\|f\|^2}$$

2. Axis:

$$l = \begin{cases} \{q + \lambda f : \lambda \in \mathbb{R}\}, & \text{if } f \neq 0 \\ \{0 + \lambda \tau : \lambda \in \mathbb{R}\}, & \text{if } f = 0 \end{cases},$$

where

$$q = \frac{\hat{f} \tau}{\|f\|^2}.$$

3. Magnitude:

$$M = \begin{cases} \|f\|, & \text{if } f \neq 0 \\ \|\tau\|, & \text{if } f = 0 \end{cases}.$$

Consider a wrench F applied to a rigid body, where

$$F = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

In another question (which, perhaps, you may have already answered) you need to report on the axis. In this question you need to calculate and fill in the pitch of the screw corresponding to the wrench F . Give your answer accurate to four significant digits.

Answer: 

The correct answer is: 0

Question 2

Correct

Mark 3.00 out of 3.00

Indicate which option regarding the computer torque control technique is false.

- ☐ a. The computed torque control technique ensures exponential trajectory tracking if the gain matrices K_p and K_v are positive definite and symmetric.
- ☐ b. The use of the computed torque control technique is supported by good experimental results.
- ☐ c. The computed torque control technique is an example of feedback linearization.
- ☒ d. The computed torque control technique uses the control law ✓

$$\tau = -K_v \dot{e} - K_p e ,$$

where τ is the joint torque vector and $e = \theta(t) - \theta_d(t)$, with $\theta(t)$ and $\theta_d(t)$ the joint angles and the desired joint angles at time t , respectively.

- ☐ e. The computed torque control technique results in joint torques that can be clearly split into feedforward and a feedback parts.

Your answer is correct.

The control law

$$\tau = -K_v \dot{e} - K_p e ,$$

where τ is the joint torque vector and $e = \theta(t) - \theta_d(t)$, with $\theta(t)$ and $\theta_d(t)$ the joint angles and the desired joint angles at time t , respectively, is the control law that is applied in PD control (see MLS section 5.3 of chapter 4, p. 193). (Any student who seriously took part in the last homework assignment of the course should have been able to answer this question without any reference to the textbook.)

The correct answer is:

The computed torque control technique uses the control law

$$\tau = -K_v \dot{e} - K_p e ,$$

where τ is the joint torque vector and $e = \theta(t) - \theta_d(t)$, with $\theta(t)$ and $\theta_d(t)$ the joint angles and the desired joint angles at time t , respectively.

Question 3

Correct

Mark 3.00 out of 3.00

Indicate which statement is false.

- ☐ a. The configuration of the tool frame of an open chain manipulator can be calculated at any time t as
$$g_{st}(\theta) = e^{\hat{\xi}_1 \theta_1} e^{\hat{\xi}_2 \theta_2} \dots e^{\hat{\xi}_n \theta_n},$$
 on condition that the base frame is chosen to be located and orientated in a very specific way.
- ☐ b. The choice of the reference configuration of an open chain manipulator is arbitrary, but the twist coordinates of the individual joints depend on this choice.
- ☐ c. The configuration of the tool frame of an open chain manipulator can be calculated at any time t as
$$g_{st}(\theta) = g_{st_1}(\theta_1) g_{l_1 l_2}(\theta_2) \dots g_{l_{n-1} l_n}(\theta_n) g_{l_n t},$$
 where $g_{l_{i-1} l_i}(\theta_i)$ is the transformation between the adjacent link frames, and $\theta(t) \in \mathbb{R}^n$ is the trajectory of joint angles in the joint angle space.
- ☒ d. The configuration of the tool frame of an open chain manipulator can be calculated at any time t as
$$g_{st}(\theta) = e^{\hat{\xi}_1 \theta_1} e^{\hat{\xi}_2 \theta_2} \dots e^{\hat{\xi}_n \theta_n} g_{st}(0),$$
 where $g_{st}(0)$ is the reference configuration of the end effector, on condition that joint n is actuated first, then the joint $n - 1$, etc., until lastly joint 1 is actuated. ✓
- ☐ e. When using the product of exponentials formula, all vectors and points are specified relative to the base coordinate frame.

Your answer is correct.

When calculating the configuration of the end effector of an open chain manipulator with the product of exponentials formula,

$$g_{st}(\theta) = e^{\hat{\xi}_1 \theta_1} e^{\hat{\xi}_2 \theta_2} \dots e^{\hat{\xi}_n \theta_n} g_{st}(0),$$

$g_{st}(\theta)$ gives the configuration of the tool frame independently of the order in which the rotations and translations are actually performed.

The correct answer is:

The configuration of the tool frame of an open chain manipulator can be calculated at any time t as

$$g_{st}(\theta) = e^{\hat{\xi}_1 \theta_1} e^{\hat{\xi}_2 \theta_2} \dots e^{\hat{\xi}_n \theta_n} g_{st}(0),$$

where $g_{st}(0)$ is the reference configuration of the end effector, on condition that joint n is actuated first, then the joint $n - 1$, etc., until lastly joint 1 is actuated.

Question 4

Correct

Mark 3.00 out of 3.00

For the vector

$$a = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

the matrix $\hat{a}$ is defined as

$$\begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix}.$$

Using Poincot's theorem, the screw coordinates of a wrench $F = (f, \tau) \in \mathbb{R}^6$ are defined as:

1. Pitch:

$$h = \frac{f^T \tau}{\|f\|^2}$$

2. Axis:

$$l = \begin{cases} \{q + \lambda f : \lambda \in \mathbb{R}\}, & \text{if } f \neq 0 \\ \{0 + \lambda \tau : \lambda \in \mathbb{R}\}, & \text{if } f = 0 \end{cases},$$

where

$$q = \frac{\hat{f} \tau}{\|f\|^2}.$$

3. Magnitude:

$$M = \begin{cases} \|f\|, & \text{if } f \neq 0 \\ \|\tau\|, & \text{if } f = 0 \end{cases}.$$

Consider a wrench F applied to a rigid body, where

$$F = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

In another question (which, perhaps, you may have already answered) you need to report on the pitch. To determine the axis of the screw, one needs to know q . In this question you need to calculate and fill in the second component (the y component) of q of the screw corresponding to the wrench F . Give your answer accurate to four significant digits.

Answer: 

The correct answer is: -1

Question 5

Correct

Mark 3.00 out of 3.00

Indicate which statement is false.

- ☐ a. If $\hat{\xi} \in se(3)$ is a twist with twist coordinates $\xi \in \mathbb{R}^6$, then, for any $g \in SE(3)$, $g\hat{\xi}g^{-1}$ is a twist with twist coordinates $Ad_g \xi \in \mathbb{R}^6$.
- ☐ b. The trajectory of a rigid body as it moves in 3D space is described by $g_{sb}(t) \in SE(3)$, in terms of a spatial frame S . Both the body velocity $V^b \in \mathbb{R}^6$ and the spatial velocity $V^s \in \mathbb{R}^6$ of the body can be calculated by calculating the time derivative of $g_{sb}(t)$ and the inverse of $g_{sb}(t)$, and combining them in a matrix product in some way.
- ☐ c. The trajectory of a rigid body as it moves in 3D space is described by $g_{sb}(t) \in SE(3)$, in terms of a spatial frame S . If, at a certain instant in time, the body velocity $V^b \in \mathbb{R}^6$ of the body is known, the spatial velocity $V^s \in \mathbb{R}^6$ of the body can be calculated by a simple matrix product with one of the factors the adjoint transformation associated with g_{sb} , and the other factor V^b .
- ☒ d. The trajectory of a rigid body as it moves in 3D space is described by $g_{sb}(t) = (p_{sb}(t), R_{sb}(t)) \in SE(3)$, in terms of a spatial frame S . If the adjoint transformation associated with g_{sb} is defined as

$$\begin{bmatrix} R_{sb} & \hat{p}_{sb}R_{sb} \\ 0 & R_{sb} \end{bmatrix},$$

then its inverse is

$$\begin{bmatrix} R_{sb}^T & R_{sb}^T \hat{p}_{sb} \\ 0 & R_{sb}^T \end{bmatrix}.$$

Your answer is correct.

The trajectory of a rigid body as it moves in 3D space is described by $g_{sb}(t) = (p_{sb}(t), R_{sb}(t)) \in SE(3)$, in terms of a spatial frame S . If the adjoint transformation associated with g_{sb} is defined as

$$\begin{bmatrix} R_{sb} & \hat{p}_{sb}R_{sb} \\ 0 & R_{sb} \end{bmatrix},$$

then its inverse is

$$\begin{bmatrix} R_{sb}^T & -R_{sb}^T \hat{p}_{sb} \\ 0 & R_{sb}^T \end{bmatrix}.$$

The correct answer is:

The trajectory of a rigid body as it moves in 3D space is described by $g_{sb}(t) = (p_{sb}(t), R_{sb}(t)) \in SE(3)$, in terms of a spatial frame S . If the adjoint transformation associated with g_{sb} is defined as

$$\begin{bmatrix} R_{sb} & \hat{p}_{sb}R_{sb} \\ 0 & R_{sb} \end{bmatrix},$$

then its inverse is

$$\begin{bmatrix} R_{sb}^T & R_{sb}^T \hat{p}_{sb} \\ 0 & R_{sb}^T \end{bmatrix}.$$

◀ Exam B Pre-registration of 036026 - Kine., Dyn. Control of Robots - Spring 2025

Jump to...

Exam A: various files. ▶

